

National Coverage Determination (NCD)

Levocarnitine for use in the Treatment of Carnitine Deficiency in ESRD Patients

230.19

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Tracking Information

Publication Number

100-3

Manual Section Number

230.19

Manual Section Title

Levocarnitine for use in the Treatment of Carnitine Deficiency in ESRD Patients

Version Number

1

Effective Date of this Version

01/01/2003

Ending Effective Date of this Version

06/19/2006

Implementation Date

01/01/2003

Description Information

Benefit Category

Drugs and Biologicals

Please Note: This may not be an exhaustive list of all applicable Medicare benefit categories for this item or service.

Item/Service Description

Carnitine is a naturally occurring substance that functions in the transport of long-chain fatty acids for energy production by the body. Deficiency can occur due to a congenital defect in synthesis or utilization, or from dialysis. The causes of carnitine deficiency in hemodialysis patients include dialytic loss, reduced renal synthesis and reduced dietary intake.

Indications and Limitations of Coverage

Intravenous levocarnitine, for one of the following indications, will only be covered for those ESRD patients who have been on dialysis for a minimum of three months.

Patients must have documented carnitine deficiency, defined as a plasma free carnitine level <40 micromol/L (determined by a professionally accepted method as recognized in current literature), along with signs and symptoms of:

- Erythropoietin-resistant anemia (persistent hematocrit <30% with treatment) that has not responded to standard erythropoietin dosage (that which is considered clinically appropriate to treat the particular patient) with iron replacement, and for which other causes have been investigated and adequately treated, or
- Hypotension on hemodialysis that interferes with delivery of the intended dialysis despite application of usual measures deemed appropriate (e.g., fluid management). Such episodes of hypotension must have occurred during at least 2 dialysis treatments in a 30-day period.

Continued use of levocarnitine will not be covered if improvement has not been demonstrated within 6 months of initiation of treatment. All other indications for levocarnitine are non-covered in the ESRD population.

For a patient currently receiving intravenous levocarnitine, Medicare will cover continued treatment if:

- Levocarnitine has been administered to treat erythropoietin-resistant anemia (persistent hematocrit <30 percent with treatment) that has not responded to standard erythropoietin dosage (that which is considered clinically appropriate to treat the particular patient) with iron replacement, and for which other causes have been investigated and adequately treated, or hypotension on hemodialysis that interferes with delivery of the intended dialysis despite application of usual measures deemed appropriate (e.g., fluid management) and such episodes of hypotension occur during at least 2 dialysis treatments in a 30-day period; and
- The patient's medical record documents a pre-dialysis plasma free carnitine level <40 micromol/L prior to the initiation of treatment; or

- The treating physician certifies (documents in the medical record) that in his/her judgment, if treatment with levocarnitine is discontinued, the patient's pre-dialysis carnitine level would fall below 40 micromol/L and the patient would have recurrent erythropoietin-resistant-anemia or intradialytic hypotension.

Claims Processing Instructions

TN AB-02-165 (Program Memorandum Intermediaries/Carriers) [↗](#)

Transmittal Information

Transmittal Number

162

Coverage Transmittal Link

<https://www.cms.gov/Regulations-and-Guidance/Guidance/Transmittals/Downloads/R162CIM.pdf> [↗](#)

Revision History

11/2002 - Implemented National Coverage Determination for Levocarnitine for End Stage Renal Disease under §1862(a)(1)(A) of the Social Security Act. Effective and implementation dates 01/01/2003. ([TN 162](#) [↗](#)) (CR 2438)

National Coverage Analyses (NCAs)

This NCD has been or is currently being reviewed under the National Coverage Determination process. The following are existing associations with NCAs, from the National Coverage Analyses database.

- [Original Consideration for Levocarnitine for End Stage Renal Disease \(CAG-00077N\)](#) [↗](#)

Additional Information

Other Versions

Title	Version	Effective Between	
Levocarnitine for use in the Treatment of Carnitine Deficiency in ESRD Patients	2	06/19/2006 - N/A	View
Levocarnitine for use in the Treatment of Carnitine Deficiency in ESRD Patients	1	01/01/2003 - 06/19/2006	You are here